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TP Aeolian Dynamics Wind Farm, 10.8 MW, at Agia Anna Design, Manufacturing, Supply, Supervision of Erection and Commissioning of the MV Switchgear and its associated Control Equipment, Telecommunication Equipment, Load Flow & Short Circuit Study and Earthing Study of the 10.8MW Wind Farm in Ag. Anna, Larnaka, Cyprus. Project: TP Aeolian Dynamics Wind Farm, 10.8MW at Ayia Anna				
Order No: P181114016		Contract No: -	Substation: Wind Farm Agia Anna S/S	
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Approved: APr 02-01-2015		Take Over Department: -	Title: TP Aeolian – Load flow & Short circuit Calculation Study	
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Revision Index

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1. General

This document provides the necessary data concerning nominal load flow and the minimum and maximum short circuit currents at medium voltage level of the TP Aeolian Dynamics Cyprus 10.8MW wind farm (TP Aeolian Dynamics Ltd).

2. Reference Documents

1. Project specifications – “EAC Specification 14-019”, Issue 05, dated 10/06/2014.
2. Project specifications – “T.P. AEOLIAN DYNAMICS WIND PARK FOR T.P. AEOLIAN DYNAMICS LTD AT AYIA ANNA/ Electrical Installations Volume B/ Electrical Specification”, ELIOFOTOU, ZINIERIS & STASIS PARTNERS.
3. IEC 60909 – (0 to 4) – “Short-circuit currents in three-phase a.c. systems”.
4. Document – “AYIA ANNA WIND FARM LOAD FLOW.pdf”, e-mail from 05/12/2014, Papachristoforou Christakis.
5. Document – “AS-site-plan-new Model (1).pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
6. Document – “Material Submissions Ayia Anna e01.pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
7. Document – “MV CABLES LENGTH.pdf”, e-mail from 06/12/2014, Papachristoforou Christakis.
8. Document – “cable.xls”, e-mail from 05/12/2014, Papachristoforou Christakis.
9. Document – “SERVER-468RG29V9258.pdf”, e-mail from 05/12/2014, Papachristoforou Christakis.
10. Document – “SERVER-468RG29V11942.pdf”, e-mail from 05/12/2014, Papachristoforou Christakis.
11. Document – “IG-159-GB-06.pdf”, e-mail from 16/12/2014, Kairis Stamatios.
12. Document – “V100-2MW_SWITCHGEAR-SETTINGS_R00.pdf”, e-mail from 16/12/2014, Kairis Stamatios.
13. Document – “REQUIRED DATA Agia Anna Wind Farm for T P Aeolian Dynamics Ltd_SP.XLS”, e-mail from 16/12/2014, Kairis Stamatios.
14. Data for 1,8MW (Vestas V-100) – Internal Transformer & Cable Between Motor and Transformer & Transformer and Wind Generator MV Board, e-mail from 19/12/2014, Kairis Stamatios.
15. Data for Overhead Line between Psevdas S/S and TP Aeolian S/S – “line aeolian.xls”, e-mail from 22/12/2014, Papadouris Savvas.
16. Data for Minimum & Maximum Short Circuit Power and Current, e-mail from 22/12/2014, Papadouris Savvas.
17. Document – “Zinieris 9_9_14 single lines drawings for the control and protection of the MW SG.pdf”, 09/09/2014, Zinieris Kyriakos.
18. Drawing – “E-01.pdf”, Drawing No: E-01, 06/10, Rev A.
19. Drawing – “E-02.pdf”, Drawing No: E-02, 06/10, Rev -.

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20. Documents – “=AA1J00+J01.pdf, =AA1J00+J02.pdf, =AA1J00+J03.pdf, 1VB2383201_001_00.pdf, 1VB2383202_001_00.pdf & 1VB2383203_001_00.pdf”, e-mails from 22/10/2014 & 24/10/2014, ABB Calor Emag.

21. General data for NEPLAN software use – “e_10_good_reasons_to_choose_neplan.pdf”.

3. Data - Assumptions

The calculations have been performed with the aid of NEPLAN Version 5.3.4 software, which conforms to the IEC calculation standards. The short circuit calculation study has been performed according to the IEC 60909 standard and the load flow analysis according to the extended Newton Raphson method.

3.1 Data

- 1) The technical data for network feeder were elicited from documents [16] & [19] of A. Eracleous Electrical Installations Ltd calculations. The data used are shown at Annex A.
- 2) The technical data for 22kV MV cables were elicited from documents [4], [5], [6], [7], [8], [9], [10], [14], [15] & [18]. Due to the fact that zero sequence capacitance was not given for document [14] it was assumed that zero sequence capacitance is equal to positive sequence capacitance. The data used are shown at Annex A. Furthermore the abovementioned cables are buried in the ground and routed through pipes.
- 3) The technical data for the wind generator transformers were given from the manufacturers in document [14] (quoting “we confirm that the data are the same as in Koshi WF”) and information was also extracted from documents [4] & [19]. The data used are shown at Annex A.
- 4) The technical data for the 1.8MW asynchronous wind generators were drawn out of document [13]. The data used are shown in Annex A.

Further data concerning the protective devices used in the MV Switchgear panel of each wind generator were taken from documents [13], [11] & [12]. Next data concerning the MV Switchgear panel (circuit breakers, current transformers and panel busbar electrical data) at the TP Aeolian Dynamics S/S were taken from documents [20]. Furthermore other useful data were extracted from the load flow study [4], the recommended relay protection settings from Vestas [12] and the single line diagram [19]. Finally technical data concerning the substation topology, cable lengths and cable routing were drawn out from document [18].

3.2 Assumptions

- 1) Due to lack/ mismatch of technical data for the M.V. cable 3x1x300mm² Al 22kV, the electrical data given from document [15] were used and modified according to reference document [3]. The abovementioned cable was assumed to be buried in the ground and its electrical data are presented in Annex A.
- 2) Due to mismatch of technical data for M.V. cables EPR or HEPR 3x70/70mm² Cu 22kV, the values of the zero sequence resistance and inductance were calculated according to document [3] and are presented in Annex A. The aforementioned cables are assumed to be routed in air.

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- 3) Due to mismatch between the given technical data from document [14] and the values calculated automatically by NEPLAN software, as well as by hand, the simulation was made with the data presented in Annex A.
- 4) The maximum short circuit calculations were made according to standard IEC 60909 – 0/(2001) assuming that all switches and breakers were closed (no open points in the examined network). Furthermore it was assumed that all nodes have 110% of their nominal voltage at the short circuit start and the network feeder and asynchronous machines contribute to the short circuit with their maximum capacity.
- 5) The minimum short circuit calculations were made according to standard IEC 60909 – 0/(2001) assuming that all switches and breakers were closed (no open points in the examined network). Furthermore it was assumed that all nodes have 90% of their nominal voltage at the short circuit start and the network feeder contributes to the short circuit. Asynchronous machines contribution to the short circuit is neglected by NEPLAN, according to IEC standard guidelines. Additionally it has to be mentioned that the resistance of the lines (cables) are taken by increased temperature.

The technical data for cables used are shown at Annex A.

4. Results - Conclusion

The layout of installation and results are also shown on the drawings at Annex B.

P = Active power losses of the element in MW

Q = Reactive power losses the element in Mvar

I = Current flow through the element in kA

I_k" = Initial short circuit current in kA

I_k" (RST) = Initial short circuit current in the phase system in kA

S_k" = Initial short circuit power in MVA

As presented in Annex B and Table [1] below, it can be mentioned that the short circuit withstand level selected for the various MV equipment of the "T.P. Aeolian Dynamics Wind Power Plant" is significantly higher than the expected fault values. Thus it can be concluded that the selected MV equipment is adequate. Furthermore the current transformers rating selected for the incomer and feeder MV panels of the T.P. Aeolian Dynamics S/S prove adequate according to the load flow calculation.

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Table [1]. Fault Level Values for T.P. Aeolian Dynamics Wind Power Plant.

Fault Levels			
S.C Type	Value	S.C Type	Value
Minimum S.C 1- phase (kA) (REF 630 - INCOMER)	1,747	Minimum S.C 1- phase (kA) (REF 630 IPPM 1)	1,747
Maximum S.C 1- phase (kA) (REF 630 - INCOMER)	2,754	Maximum S.C 1- phase (kA) (REF 630 IPPM 1)	2,754
Minimum S.C 2- phase (kA) (REF 630 - INCOMER)	1,834	Minimum S.C 2- phase (kA) (REF 630 IPPM 1)	1,834
Maximum S.C 2- phase (kA) (REF 630 - INCOMER)	4,456	Maximum S.C 2- phase (kA) (REF 630 IPPM 1)	4,456
Minimum S.C 3- phase (kA) (REF 630 INCOMER)	2,118	Minimum S.C 3- phase (kA) (REF 630 IPPM 1)	2,118
Maximum S.C 3- phase (kA) (REF 630 INCOMER)	5,145	Maximum S.C 3- phase (kA) (REF 630 IPPM 1)	5,145
Minimum S.C 2- phase to ground (kA) (REF 630 - INCOMER)	1,978	Minimum S.C 2- phase to ground (kA) (REF 630 IPPM 1)	1,978
Maximum S.C 2- phase to ground (kA) (REF 630 - INCOMER)	4,558	Maximum S.C 2- phase to ground (kA) (REF 630 IPPM 1)	4,558
S.C Type	Value	S.C Type	Value
Minimum S.C 1- phase (kA) (REF 630 IPPM 2)	1,747	Minimum S.C 1- phase (kA) (ecorRPG202A CGMCMOS)	1,578
Maximum S.C 1- phase (kA) (REF 630 IPPM 2)	2,754	Maximum S.C 1- phase (kA) (ecorRPG202A CGMCMOS)	2,725
Minimum S.C 2- phase (kA) (REF 630 IPPM 2)	1,834	Minimum S.C 2- phase (kA) (ecorRPG202A CGMCMOS)	1,770
Maximum S.C 2- phase (kA) (REF 630 IPPM 2)	4,456	Maximum S.C 2- phase (kA) (ecorRPG202A CGMCMOS)	4,417
Minimum S.C 3- phase (kA) (REF 630 IPPM 2)	2,118	Minimum S.C 3- phase (kA) (ecorRPG202A CGMCMOS)	2,044
Maximum S.C 3- phase (kA) (REF 630 IPPM 2)	5,145	Maximum S.C 3- phase (kA) (ecorRPG202A CGMCMOS)	5,101
Minimum S.C 2- phase to ground (kA) (REF 630 IPPM 2)	1,978	Minimum S.C 2- phase to ground (kA) (ecorRPG202A CGMCMOS)	1,170
Maximum S.C 2- phase to ground (kA) (REF 630 IPPM 2)	4,558	Maximum S.C 2- phase to ground (kA) (ecorRPG202A CGMCMOS)	4,530

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5. Annex A

The data presented in this paragraph were elicited from the reference documents, presented above.

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NETWORK FEEDER (EAC NETWORK)

NAME	Sk"min	Sk"max	Ik"min	Ik"max	Z(0)/Z(1)min	Z(0)/Z(1)max	R(1)/X(1)min	R(1)/X(1)max	R(0)/X(0)min	R(0)/X(0)max
	MVA	MVA	kA	kA						
EAC NETWORK	80,783	163	2,12	4,278	1,631	2,9846	0,1	0,1	0,1	0,1

LINE DATA

NAME	TYPE	LENGTH	Number of parallel cables	R(1)	X(1)	C(1)	R(0)	X(0)	C(0)	Q	Material	Temp. end of SC	Max Oper.T emp.	Reduction factor
		km		Ohm/km	Ohm/km	µF/km	Ohm/km	Ohm/km	µF/km	mm ²		°C	°C	
3x(1x300mm ²)	N2XSY 3X1X300 12/22kV	0,1	1	0,0128	0,011	0,325	0,0685	0,0275	0,325	300	Al	250	90	0,809
Cable C1	N2XSY 3x(1x150mm ²) 12/22kV	0,706	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
Cable C2	N2XSY 3x(1x150mm ²) 12/22kV	0,77	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C3	N2XSY 3x(1x150mm ²) 12/22kV	0,487	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
Cable C4	N2XSY 3x(1x150mm ²) 12/22kV	0,346	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C5	N2XSY 3x(1x150mm ²) 12/22kV	0,994	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,593
Cable C6	N2XSY 3x(1x150mm ²) 12/22kV	0,542	1	0,269	0,00039	0,249	1,54944	0,00109624	0,249	150	Al	250	90	0,688
CBL WG_1	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_2	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_3	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_4	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_5	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473
CBL WG_6	EPR 3x70/70mm ² 22kV	0,095	1	0,347	0,099	0,325	1,041	0,297	0,325	70	Cu	250	120	0,473

TRANSFORMERS DATA

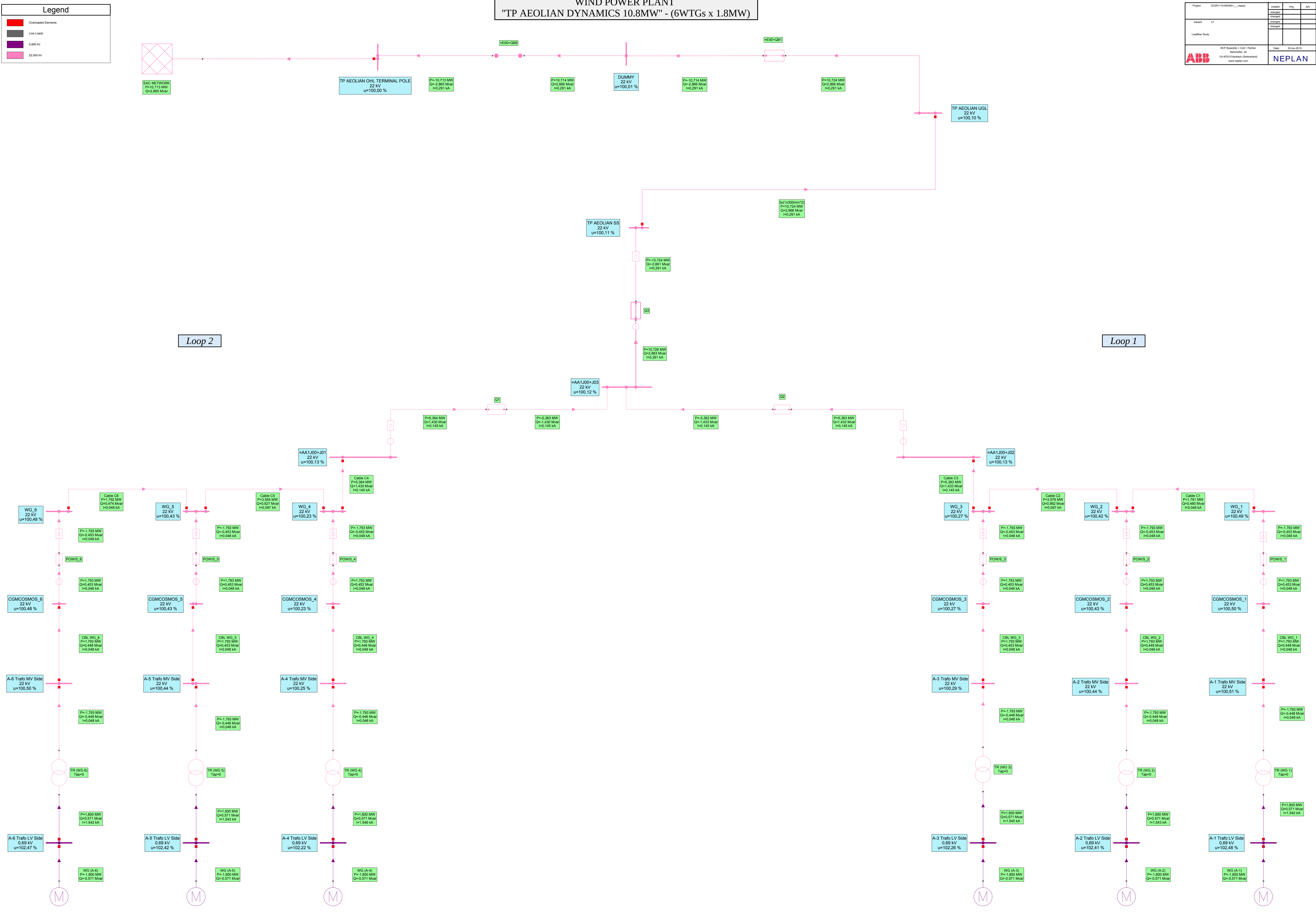
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		MVA	kV	kV	%	%	%	%	%	kW		
TR (WG 1)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 2)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 3)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 4)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 5)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct
TR (WG 6)	DYn5	2,1	22	0,69	7,6	0,1857	0,594	0,1857	0	3,9	na	direct

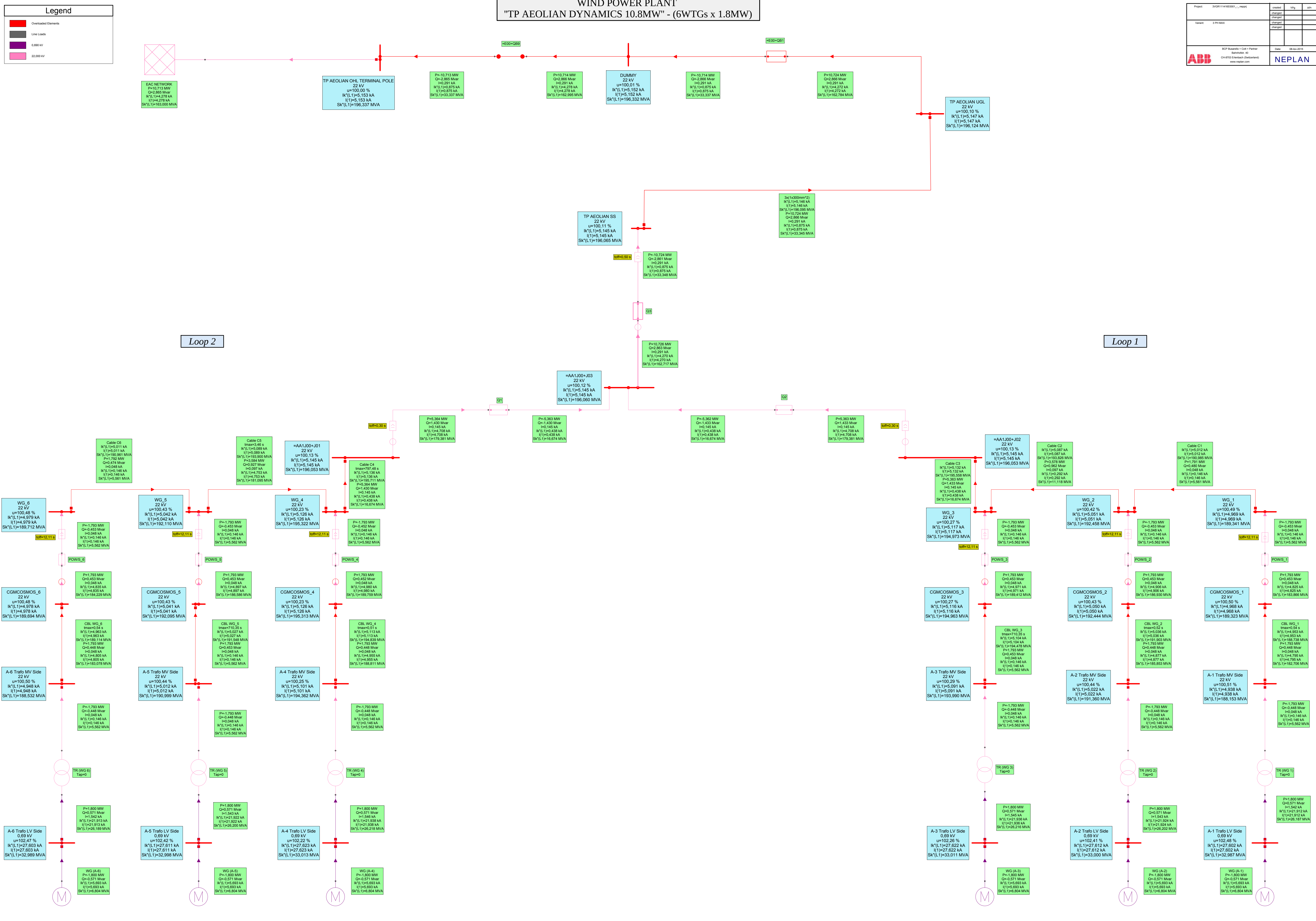
ASYNCHRONOUS DOUBLE FED WIND GENERATORS

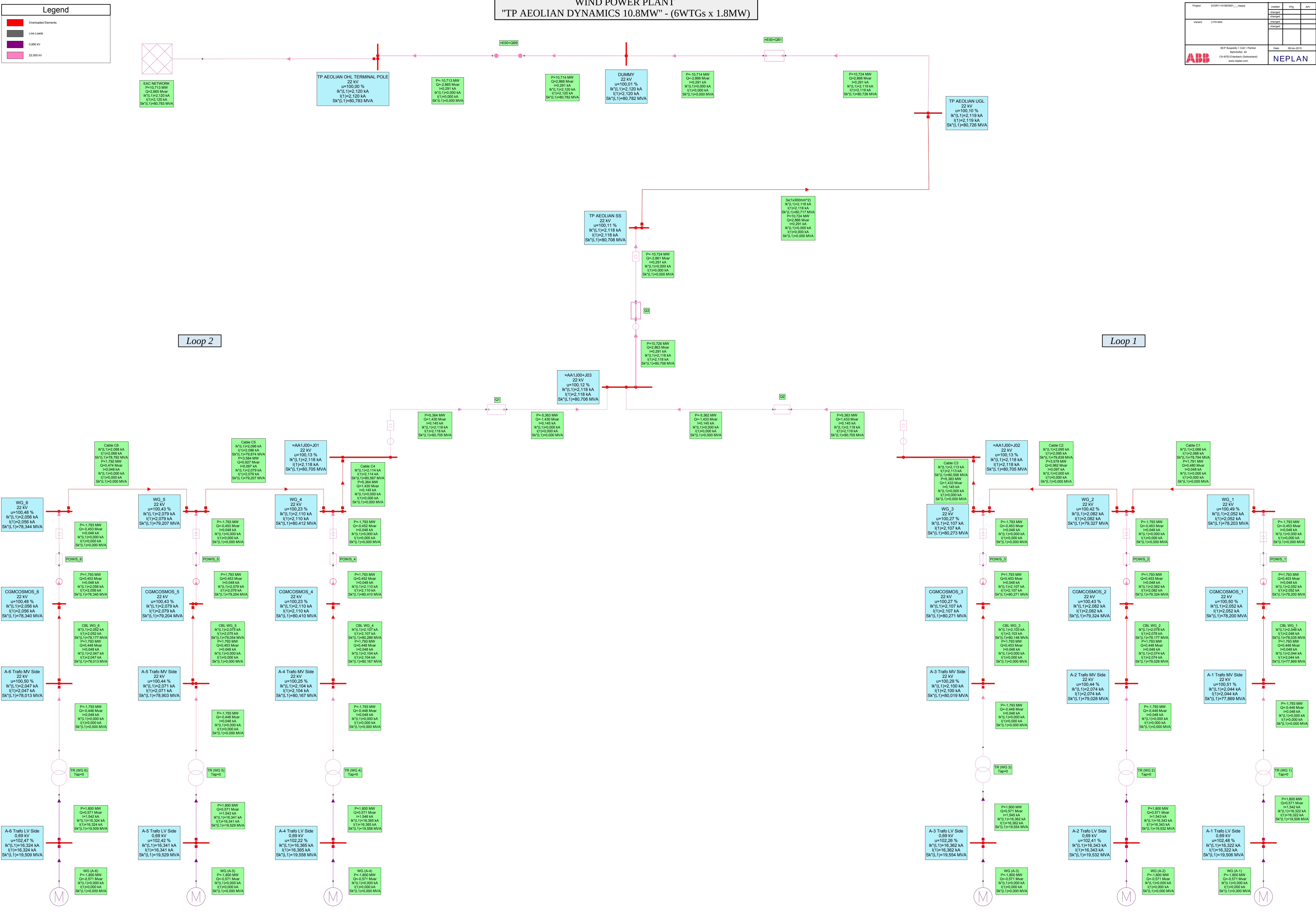
NAME	Pr	Sr	Ur	Ir	cosphi	eta	Ia/Ir	pole pairs	cosphi start	Ma/Mr	Mk/Mr	Rm	sr	Slip
	MW	MVA	kV	kA	-	-	-	-	-	-	-	Ohm	%	%
WG (A-1)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-2)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-3)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-4)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-5)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12
WG (A-6)	1,8	2,062	0,69	1,7252	0,9	0,97	6,474	2	0,2	0,2917	3,0722	0,007173	0,6	-12

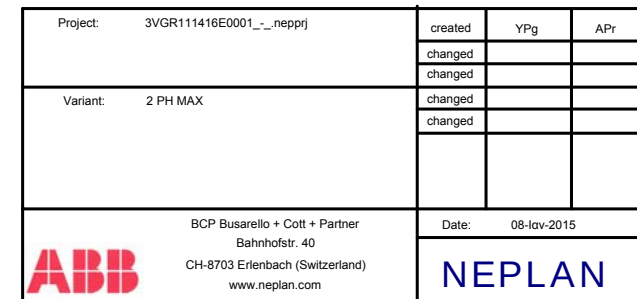
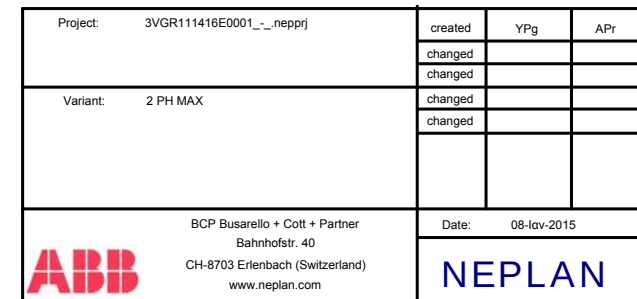
6. Annex B

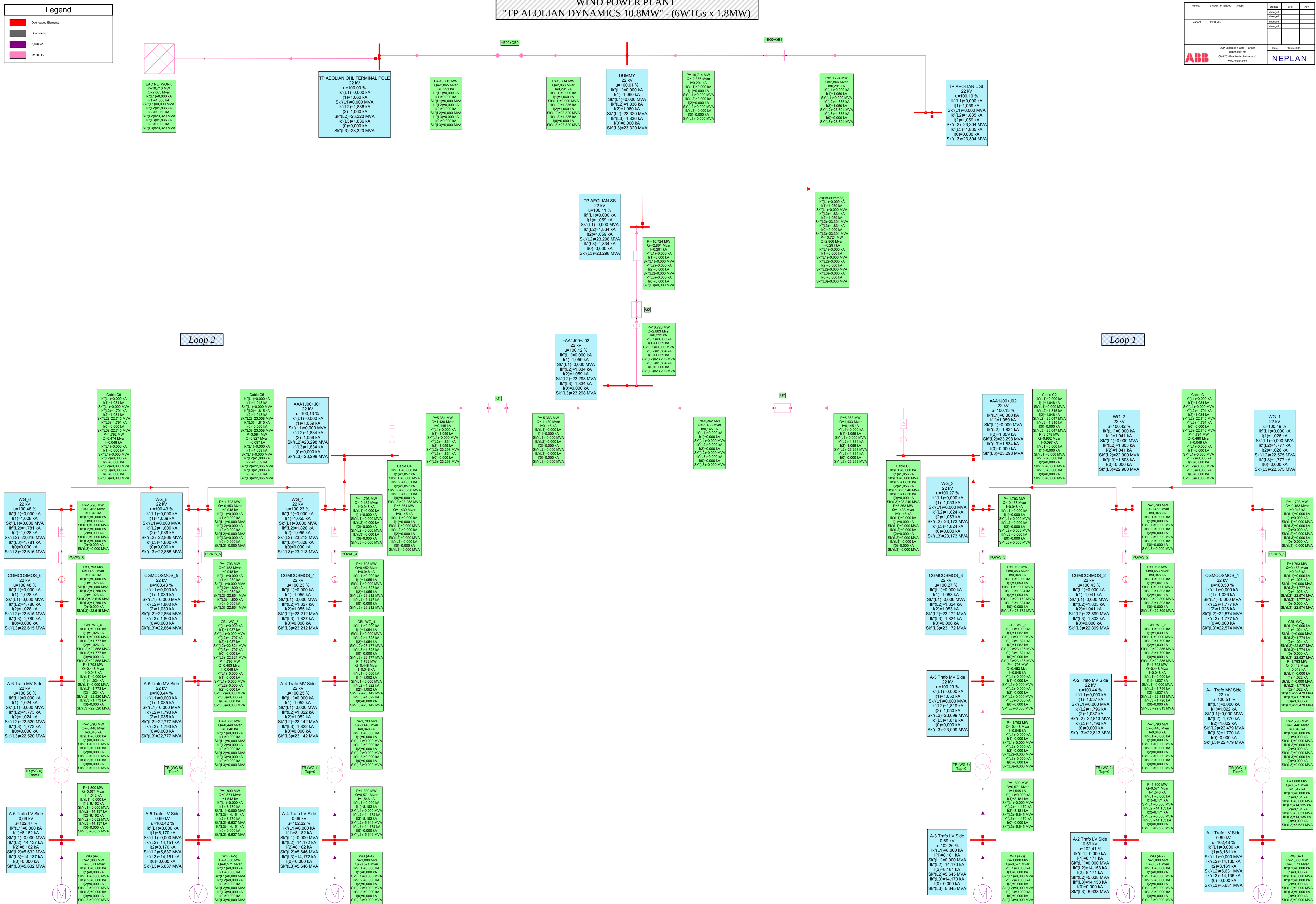
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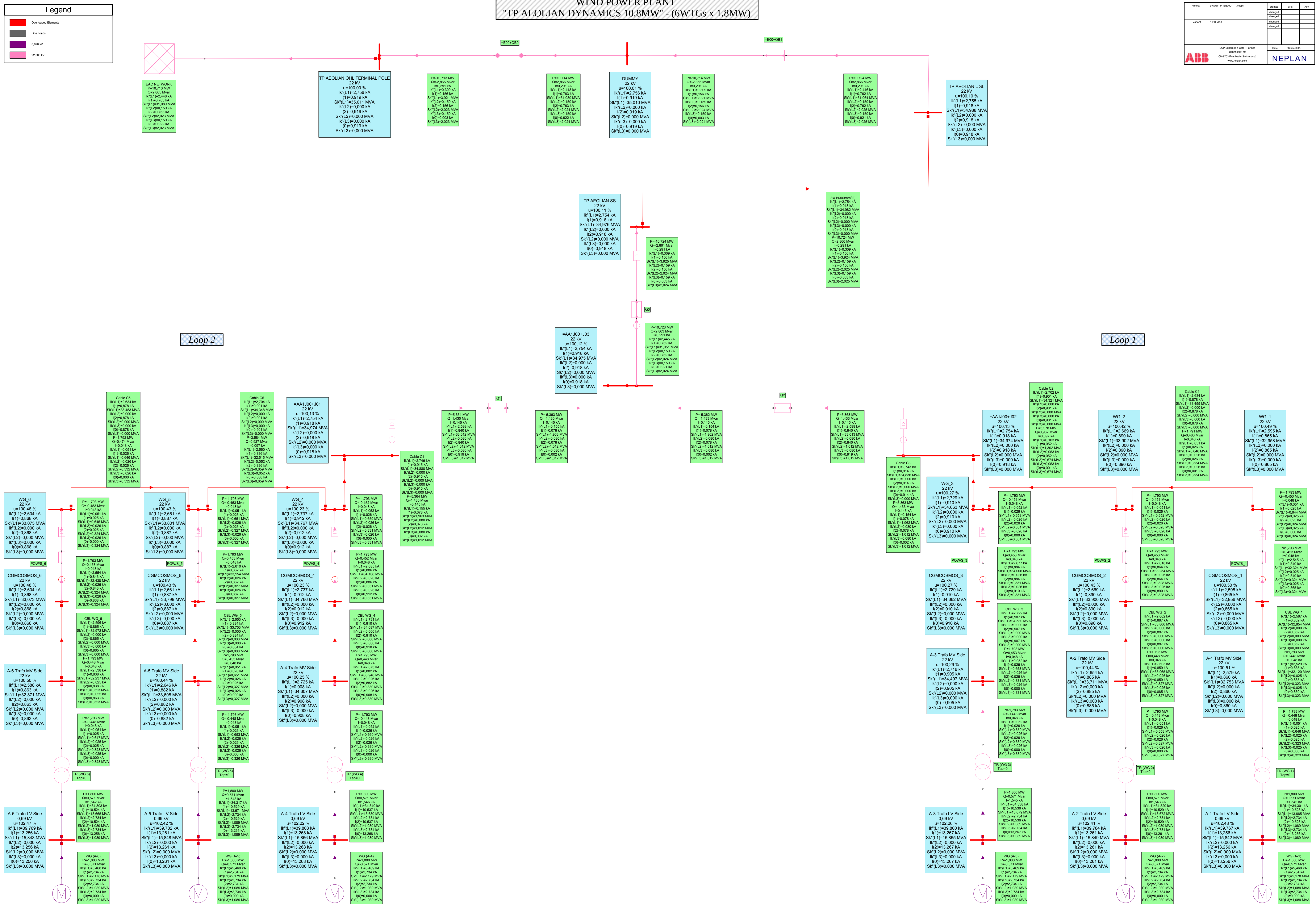


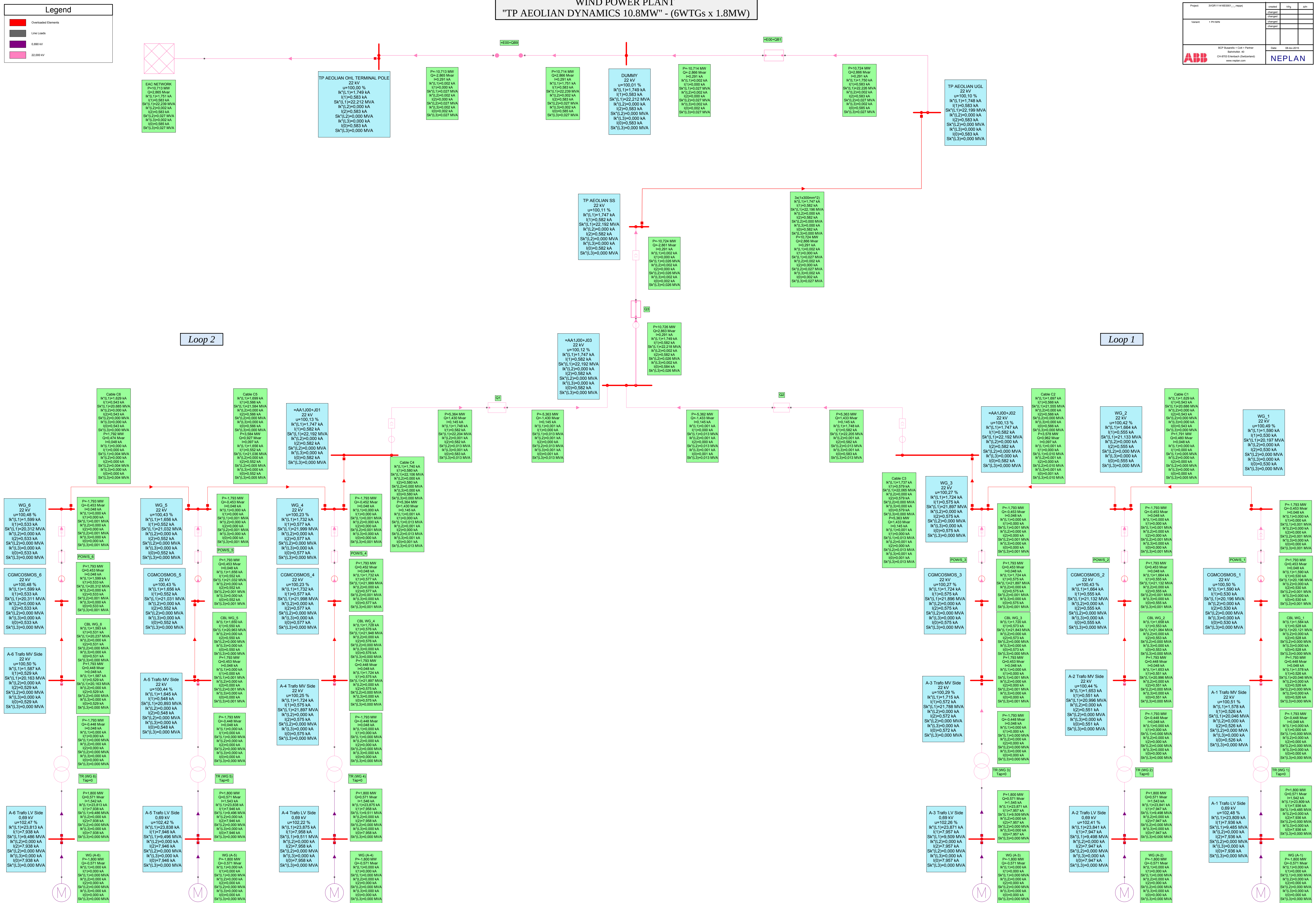


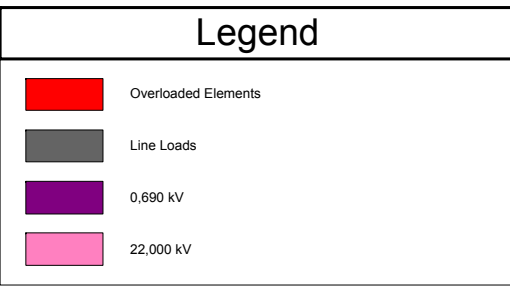












WIND POWER PLANT
"TP AEOLIAN DYNAMICS 10.8MW" - (6WTGs x 1.8MW)

Project	3VGR114480001_1_neppj	created	YPlg	Abn
Variant	2 PH TO GROUND MAX	changed		
		changed		
		changed		
	BCP Busarelio + Cof + Partner Bahnhofstr. 40 CH-8703 Erlenbach (Switzerland) www.neplan.com	Date	09-jan-2015	
				

